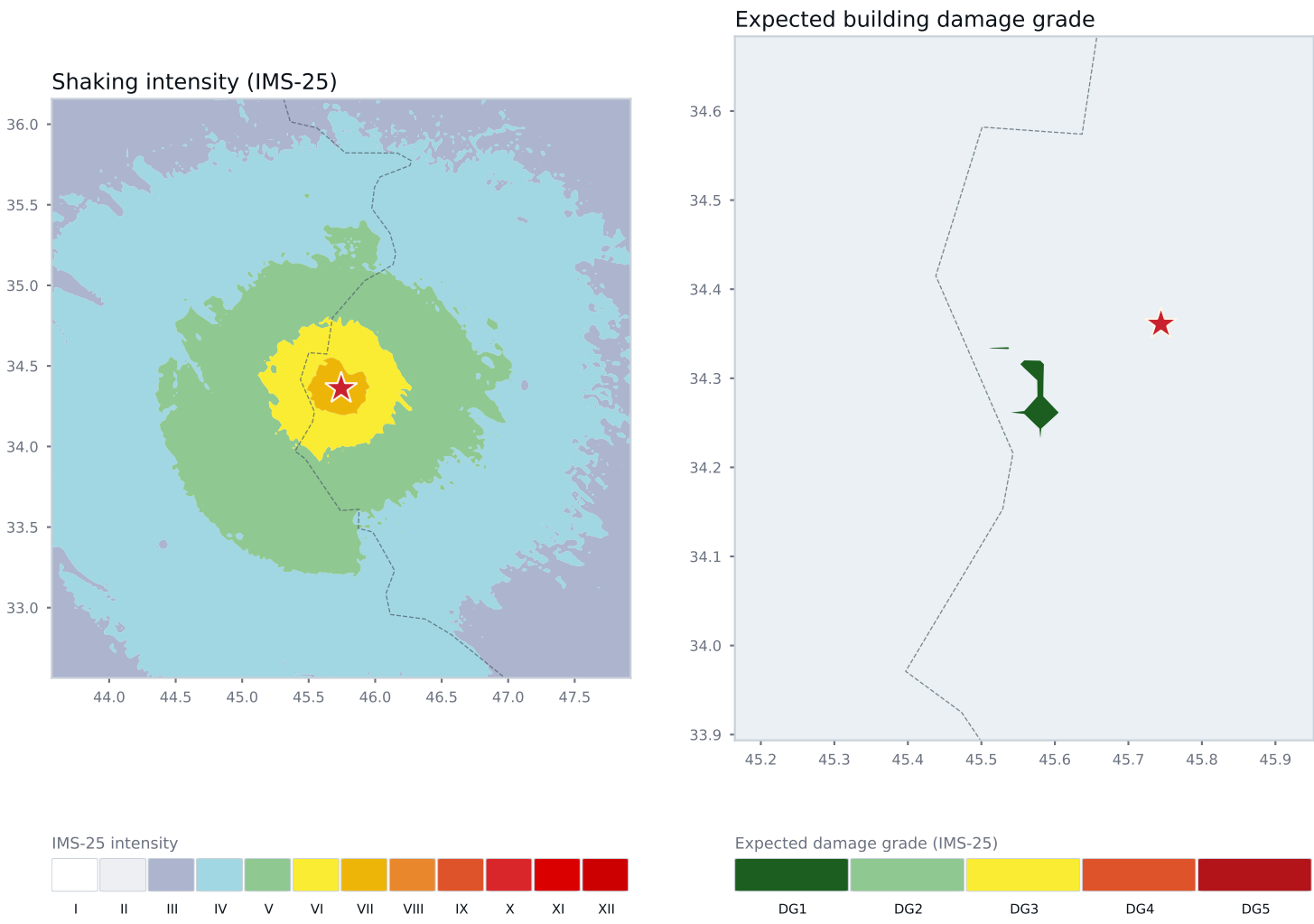




Damage alert

ORANGE

Widespread damage; many buildings unusable

Provisional

Transit occupancy

Buildings heavily damaged (DG3 or worse)

1.2 thousand to 15 thousand 5.3 million

Central estimate 3.7 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

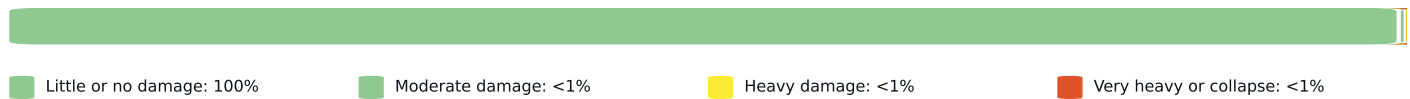
21% of them inferred from built-up area, not observed footprints.

People by shaking level



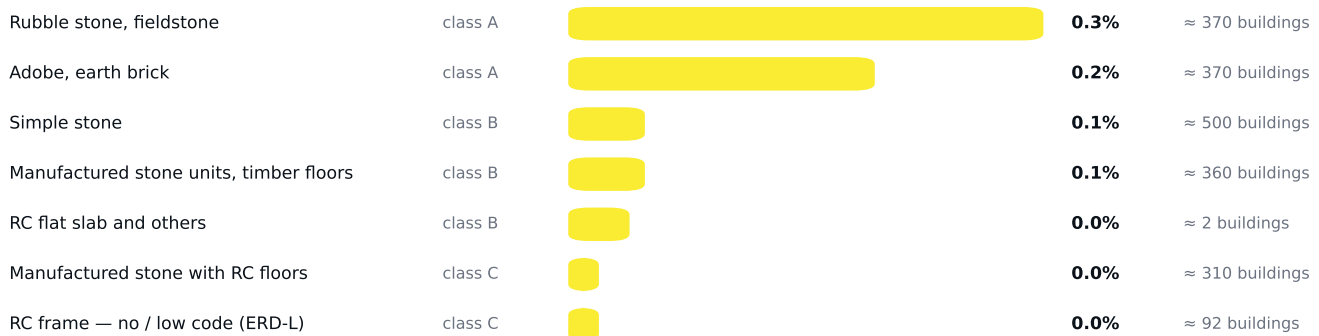
Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage

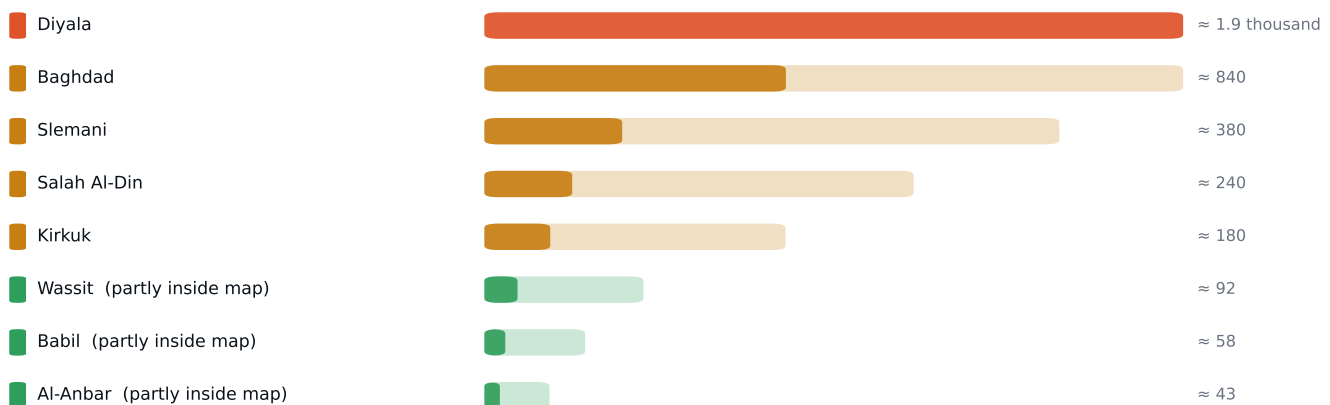


Which buildings failed

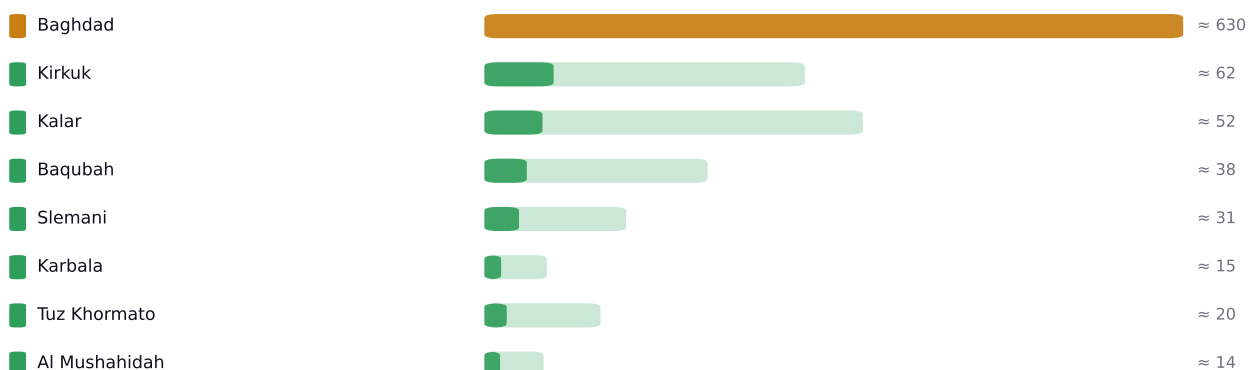
Share of each type's own buildings heavily damaged, worst first



Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Diyala	602 thousand	980	520 to 7.8 thousand	0.16%
Baghdad	2.1 million	470	280 to 3.0 thousand	0.02%
Slemani	588 thousand	220	120 to 1.6 thousand	0.04%
Salah Al-Din	411 thousand	140	67 to 1.2 thousand	0.03%
Kirkuk	549 thousand	120	69 to 840	0.02%
Wassit	195 thousand	51	28 to 440	0.03%
Babil	219 thousand	40	23 to 280	0.02%
Al-Anbar	172 thousand	29	15 to 180	0.02%
Kerbala	201 thousand	21	10 to 120	0.01%
Erbil	255 thousand	20	10 to 100	0.01%
Ninewa	720	none	none to 1	0%
Maysan	660	none	none to 1	0%

How this report was made

Hazard engine	Bumelerze SHAKEmap 26.1.1, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)
Conditioning	mvn (Engler et al. 2022) on dyfi with reverse-GMICE sigma (UQ-B, D46)
Observations	0 seismic stations, 122 felt-report cells (felt channel: dyfi)
Distances	ps2ff; rupture: Origin
Damage scale	IMS-25
Provenance	config 09a6a04fa72f, product version 1, event bml20180002

Data sources

Earthquake parameters	USGS, EMSC, GEOFON
Building footprints	GFZ OpenBuildingMap, from OpenStreetMap (c) OpenStreetMap contributors, ODbL; Microsoft Building Footprints, CDLA-Permissive-2.0
Built-up surface, settlements	European Commission JRC, Global Human Settlement Layer, CC BY 4.0
Population	WorldPop, University of Southampton, CC BY 4.0
Administrative boundaries	OCHA COD-AB Iraq, CC BY-IGO; source authority Iraq Central Statistical Organisation
Building types, vulnerability	GEM Global Exposure Model; IMS-25 vulnerability table
Casualty rates (computed, not published)	FEMA Hazus Earthquake Model Technical Manual 5.1

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Bumelerze SHAKEmap and damage report, event bml20180002, version 1, generated 21 Sep 2026, 22:36 UTC. CC BY 4.0.